# Mehran Aghabozorgi

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## Research Interests

**Generative Modeling**: efficient generation, sample efficiency, multimodal generation
**Reinforcement Learning**: world models, model-based RL, offline RL, sample-efficient continuous control
**Multimodal & Robot Learning**: vision-language-action (VLA) policies, RL for VLAs, efficient inference for large multimodal / transformer models

## Education

**Simon Fraser University** — Vancouver, Canada
*Ph.D. in Computer Science — APEX Lab (Advisor: Ke Li); expected Fall 2026* — *2021 – present*
- Affiliated with the Alberta Machine Intelligence Institute (Amii).

**Isfahan University of Technology** — Isfahan, Iran
*B.Sc. in Computer Engineering — GPA 18.29/20 (ranked 1st)* — *2015 – 2020*

## Experience

**Mitacs Research Intern** — Jan 2026 – Sep 2026
*RBC Borealis* — *Vancouver, Canada*
- Accelerated Vision-Language-Action (VLA) policies whose inference cost limits real-time robotic control.
- Developed attention-guided token pruning and compressed-sensing token compression to cut latency and compute while preserving task success.

**Research Intern** — Aug – Oct 2019
*Max Planck Institute for Informatics* — *Saarbrücken, Germany*
- Enhanced the preprocessing phase of the SPASS SAT solver with Prof. Christoph Weidenbach; demonstrated improvements on competition-grade benchmarks.

**Senior Developer** — 2017 – 2020
*Morvarid*
- Shipped production systems in Python, Java, Scala, and JavaScript: an anonymous, verifiable voting system and a security-focused cloud for end-to-end encrypted file storage.
- Led engineering for Ergo Pool (mining backends and payout automation) on the Ergo blockchain.

## Papers

**Dynamic Episode ALignment: Sample-Efficient Reinforcement Learning of Vision-Language-Action Models** — 2027
**Mehran Aghabozorgi**, Kian Hosseinkhani, Haowen Yao, Luis Figueredo, Ke Li
*Under review*

**PAIR: Distance-Preserving Compression for Fast Inference and Adaptation of Vision-Language-Action Policies** — 2027
Kian Hosseinkhani*, **Mehran Aghabozorgi***, Kry Yik Chau Lui, Kaito Suzuki, Ken Jiankun Zheng, Andy Zhang, Dantong Niu, Trevor Darrell, Roei Herzig, Ke Li (*equal contribution)
*Under review*

**IMLE-LLM: Latent-Conditioned Windowed Decoding for Faster Inference** — 2027
Alireza Moazeni, **Mehran Aghabozorgi**, Tristan Engst, Ke Li
*Under review*

**WIMLE: Uncertainty-Aware World Models with IMLE for Sample-Efficient Continuous Control** — 2026
**Mehran Aghabozorgi**, Yanshu Zhang, Alireza Moazeni, Ke Li
*International Conference on Learning Representations (ICLR)* [project page]

**IMLE-VLA: Fast Single-Step Action Generation for Vision-Language-Action Policies** 2026
Kian Hosseinkhani, Qinhe Peng, George Shramko, **Mehran Aghabozorgi**, Jianing Qian, Tristan Engst, Alireza Moazeni, Dinesh Jayaraman, Ke Li
*IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* [project page]

**P-CORE: Self-Supervised Surface Consistency for Point-Based Neural Editing** 2026
Yanshu Zhang, Shichong Peng, **Mehran Aghabozorgi**, Alireza Moazeni, Ke Li
*European Conference on Computer Vision (ECCV)* [project page]

**Adaptive IMLE for Few-shot Pretraining-free Generative Modelling** 2023
**Mehran Aghabozorgi**, Shichong Peng, Ke Li
*International Conference on Machine Learning (ICML)* [project page]

## Technical Skills

**Programming**: Python, C++, Scala, Java, JavaScript
**ML & Deep Learning**: PyTorch, JAX, distributed training (DDP, FSDP)
**RL & Simulation**: MuJoCo, DeepMind Control, MyoSuite

## Awards & Scholarships

- Helmut & Hugo Eppich Family Graduate Scholarship, Simon Fraser University (2026).
- Ph.D. Research Scholarship, Simon Fraser University — multiple terms (2024–2025).
- Graduate Fellowship, Simon Fraser University — multiple terms (2021–2024).
- ACM-ICPC West Asia Regional — **8th** (40th, Team TRIO) and **9th** (41st, Team Valar Geekolous).
- Ranked **2nd**, nationwide ICPC competition 2017 (Team Valar Geekolous).
- Nationwide AI Challenge (UIAI) — ranked **1st** (2018) and **2nd** (2017).